

Model Lab

Every prior, same slate — 2026-09-25

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

37,579

Held-Out Scores

Fitted on plate appearances before 2026-08-18, scored on the 37,579 that came after (actual HR rate 2.94%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5757	1.48×	0.02846	0.13171	0.00332
KNN comps Prior is the pooled rate of the nearest comparables.	0.5890	1.48×	0.02847	0.13135	0.00564
SVR Support-vector regression on the same feature matrix.	0.5817	1.46×	0.02857	0.13296	0.00857
Random forest Forest regression; a learned similarity metric.	0.5822	1.43×	0.02853	0.13216	0.00776
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5890	1.48×	0.02847	0.13135	0.00564
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5886	1.49×	0.02847	0.13139	0.00574
Linear regression Ordinary least squares on the same features.	0.5842	1.42×	0.02852	0.13220	0.00736
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5861	1.45×	0.02859	0.13228	0.00837
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5859	1.47×	0.02850	0.13178	0.00672
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5833	1.53×	0.02858	0.13258	0.00822

Blend weights (forest variant): KNN 0.90, SVR 0.00, RF 0.10

Shrinkage K per prior: league 253, knn 253, svr 253, rf 253, core 253, ensemble 253, linear 253, logistic 253, nn 253, xgb 253 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,350 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5259	1.23×	0.02381	0.11516	0.00818
KNN comps Prior is the pooled rate of the nearest comparables.	0.5514	0.75×	0.02402	0.11722	0.01801
SVR Support-vector regression on the same feature matrix.	0.5574	0.85×	0.02460	0.12598	0.02960
Random forest Forest regression; a learned similarity metric.	0.5581	0.85×	0.02433	0.12105	0.02511
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5514	0.75×	0.02402	0.11722	0.01801
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5533	1.04×	0.02405	0.11745	0.01820
Linear regression Ordinary least squares on the same features.	0.5513	0.85×	0.02441	0.12335	0.02541
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5515	0.66×	0.02481	0.12298	0.03277
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5606	1.04×	0.02412	0.11895	0.02007
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5695	0.94×	0.02469	0.12343	0.03050

Slate Agreement

Flat league · 3/6 shared 0/6 hit

Yordan Alvarez

Jeremy Peña

Ben Rice

Jackson Chourio

Pete Alonso

Isaac Paredes

SVR · 2/6 shared 1/6 hit

Yordan Alvarez

Emmanuel Rodriguez

Ben Rice

Kyle Schwarber

Rafael Flores Jr. ✓

Shea Langeliers

Core (KNN+SVR) · 6/6 shared 1/6 hit

Yordan Alvarez

Ben Rice

Leo Bernal

Pete Alonso

Munetaka Murakami ✓

Zack Gelof

Linear regression · 5/6 shared 0/6 hit

Yordan Alvarez

Ben Rice

Zack Gelof

Pete Alonso

Leo Bernal

Kyle Schwarber

Neural net (torch) · 5/6 shared 0/6 hit

KNN comps · 6/6 shared 1/6 hit

16.1% Yordan Alvarez

15.6% Ben Rice

15.6% Leo Bernal

15.2% Pete Alonso

15.2% Munetaka Murakami ✓

14.7% Zack Gelof

Random forest · 5/6 shared 1/6 hit

20.7% Yordan Alvarez

19.3% Ben Rice

19.0% Leo Bernal

17.9% Kyle Schwarber

17.8% Munetaka Murakami ✓

17.8% Pete Alonso

Forest blend · 6/6 shared 1/6 hit

18.8% Yordan Alvarez

17.8% Ben Rice

17.6% Leo Bernal

17.4% Pete Alonso

17.1% Munetaka Murakami ✓

16.8% Zack Gelof

Logistic regression · 4/6 shared 0/6 hit

19.6% Yordan Alvarez

17.8% Kyle Schwarber

17.3% Emmanuel Rodriguez

17.3% Ben Rice

17.3% Pete Alonso

17.2% Zack Gelof

18.8%

17.8%

17.6%

17.4%

17.1%

16.8%

20.0%

18.3%

17.8%

17.5%

17.5%

17.4%

18.9%

17.9%

17.6%

17.4%

17.1%

16.8%

21.1%

19.2%

18.6%

18.1%

17.9%

17.5%

Leo Bernal

Munetaka Murakami ✓

17.5%

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Kyle Schwarber Philadelphia Phillies · vs LHP · 45 HR / 687 PA	14.1%	16.5%	17.9%	17.5%	16.5%	16.6%	17.2%	19.2%	17.5%
9. Ben Rice slate New York Yankees · vs RHP · 41 HR / 658 PA	14.3%	16.4%	17.4%	16.8%	16.4%	16.4%	16.4%	16.7%	16.8%
10. Shea Langeliers Athletics · vs LHP · 23 HR / 456 PA	13.2%	16.2%	17.8%	16.5%	16.2%	16.2%	15.7%	17.3%	17.0%
11. Pete Crow-Armstrong Chicago Cubs · vs RHP · 45 HR / 716 PA	14.7%	16.1%	17.1%	16.7%	16.1%	16.2%	16.2%	17.3%	16.8%
12. Jackson Chourio Milwaukee Brewers · vs LHP · 25 HR / 563 PA	15.2%	16.0%	16.7%	16.8%	16.0%	16.1%	16.2%	16.4%	16.6%
13. Rafael Flores Jr. Pittsburgh Pirates · vs RHP · 10 HR / 155 PA	12.8%	16.0%	17.8%	16.3%	16.0%	16.0%	16.7%	16.4%	17.1%
14. Pete Alonso slate Baltimore Orioles · vs LHP · 41 HR / 690 PA	13.1%	16.0%	16.2%	15.9%	16.0%	16.0%	15.8%	16.5%	15.9%
15. Jeremy Peña Houston Astros · vs RHP · 20 HR / 466 PA	15.6%	15.9%	16.5%	16.3%	15.9%	16.0%	15.8%	15.8%	15.9%
16. Brandon Lowe Pittsburgh Pirates · vs RHP · 34 HR / 646 PA	13.9%	15.7%	16.0%	16.1%	15.7%	15.7%	15.8%	16.8%	16.3%
17. Miguel Vargas Chicago White Sox · vs RHP · 33 HR / 695 PA	13.0%	15.6%	15.4%	15.0%	15.6%	15.5%	15.4%	15.9%	15.7%
18. Coby Mayo Baltimore Orioles · vs LHP · 24 HR / 444 PA	12.9%	15.5%	16.3%	16.1%	15.5%	15.6%	15.6%	16.9%	16.2%
19. Zach Neto Los Angeles Angels · vs LHP · 26 HR / 689 PA	13.1%	15.5%	15.2%	15.0%	15.5%	15.4%	14.5%	15.3%	14.6%
20. Kazuma Okamoto Toronto Blue Jays · vs RHP · 33 HR / 625 PA	13.7%	15.4%	16.5%	15.6%	15.4%	15.4%	15.5%	16.2%	16.1%
21. Pete Crow-Armstrong Chicago Cubs · vs LHP · 45 HR / 716 PA	13.5%	15.2%	16.3%	15.8%	15.2%	15.2%	15.3%	16.5%	15.9%
22. Luis García Jr.	13.9%	15.2%	16.7%	16.1%	15.2%	15.3%	15.4%	16.1%	16.0%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
New York Yankees · vs RHP · 30 HR / 539 PA									
23. Oneil Cruz Pittsburgh Pirates · vs RHP · 20 HR / 414 PA	12.9%	15.1%	14.5%	14.1%	15.1%	15.0%	14.7%	14.3%	14.7%
24. Mike Trout Los Angeles Angels · vs LHP · 22 HR / 624 PA	12.1%	15.1%	14.2%	14.7%	15.1%	15.0%	15.1%	15.8%	15.1%
25. Colson Montgomery Chicago White Sox · vs RHP · 31 HR / 629 PA	12.4%	14.9%	14.2%	14.5%	14.9%	14.9%	15.2%	16.5%	14.9%
26. Emmanuel Rodriguez Minnesota Twins · vs RHP · 2 HR / 44 PA	9.8%	14.8%	19.3%	16.2%	14.8%	15.0%	16.1%	18.6%	15.8%
27. Kyle Teel Chicago White Sox · vs RHP · 5 HR / 136 PA	13.5%	14.8%	14.0%	13.9%	14.8%	14.7%	14.6%	12.7%	13.2%
28. Junior Caminero Tampa Bay Rays · vs RHP · 41 HR / 692 PA	14.1%	14.7%	17.2%	14.9%	14.7%	14.7%	15.1%	15.6%	15.6%
29. Cal Raleigh Seattle Mariners · vs RHP · 23 HR / 520 PA	12.2%	14.6%	13.6%	14.1%	14.6%	14.5%	14.9%	15.8%	14.9%
30. James Wood Washington Nationals · vs LHP · 30 HR / 610 PA	11.4%	14.5%	15.2%	14.8%	14.5%	14.6%	14.0%	14.5%	14.5%
31. Griffin Conine Miami Marlins · vs RHP · 17 HR / 311 PA	12.4%	14.5%	15.7%	15.1%	14.5%	14.6%	15.1%	15.6%	15.4%
32. Andrés Chaparro Washington Nationals · vs LHP · 11 HR / 221 PA	12.8%	14.4%	14.7%	14.4%	14.4%	14.4%	13.7%	13.7%	14.3%
33. Francisco Lindor New York Mets · vs RHP · 20 HR / 454 PA	13.9%	14.3%	14.7%	14.3%	14.3%	14.3%	14.3%	14.5%	14.3%
34. Randal Grichuk Chicago White Sox · vs RHP · 14 HR / 269 PA	11.8%	14.3%	15.1%	13.5%	14.3%	14.2%	13.7%	15.5%	14.9%
35. Jordan Walker St. Louis Cardinals · vs RHP · 28 HR / 655 PA	12.9%	14.2%	15.1%	14.1%	14.2%	14.2%	14.4%	15.1%	15.0%
36. Fernando Tatis Jr. San Diego Padres · vs RHP · 25 HR / 696 PA	13.6%	14.2%	13.7%	13.5%	14.2%	14.1%	14.4%	14.6%	14.5%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Jake Bauers Milwaukee Brewers · vs LHP · 26 HR / 565 PA	12.4%	14.2%	14.4%	14.7%	14.2%	14.2%	14.0%	14.0%	14.2%
38. Matt Olson Atlanta Braves · vs LHP · 41 HR / 688 PA	12.3%	14.1%	15.3%	14.7%	14.1%	14.2%	14.3%	15.4%	14.8%
39. Bobby Witt Jr. Kansas City Royals · vs LHP · 18 HR / 619 PA	12.7%	14.1%	14.3%	14.0%	14.1%	14.1%	14.0%	14.8%	14.5%
40. Roman Anthony Boston Red Sox · vs RHP · 5 HR / 232 PA	12.5%	14.1%	12.0%	14.0%	14.1%	14.1%	13.3%	12.2%	13.2%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.